DAY 13 · THEORY HANDOUT

Inside a Transformer Block

Attention gathers context. Other parts process, preserve, and balance the numbers.

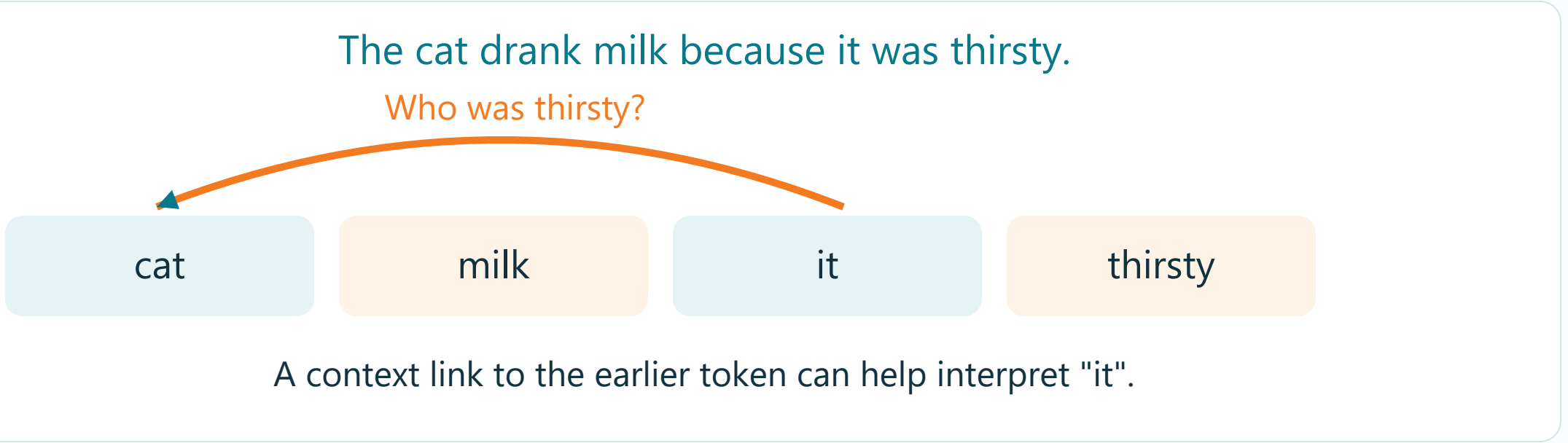
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1. Attention gathers useful context

Attention is a learned way for a token to collect information from other available tokens.



A context link to the earlier token can help interpret "it".

The model gives different amounts of influence to available tokens, then mixes their information. It updates the current token's representation.

Why use it? A nearby word is not always the useful clue. An earlier token can help explain what a later token refers to.

> The arrow shows an intuition, not measured model attention. Attention patterns do not prove a model's full reasoning.

2. Self-attention and causal masking

Self-attention mixes information from tokens in the same sequence.



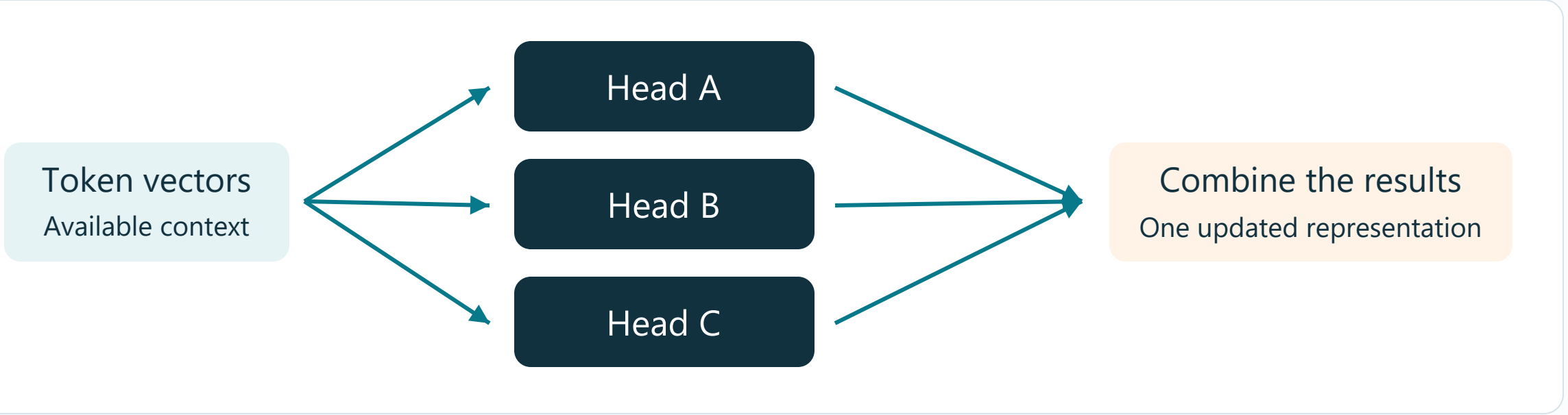
A causal mask blocks access to later positions. In a GPT-style model, each position can use itself and earlier tokens.

Why block later tokens? During training, the model must predict what comes next without seeing that answer. BERT-style models can instead use both sides for their reading tasks.

> This is attention masking, not the zero padding used to make input rows equal in length.

3. Multiple attention heads

An attention head is one learned view of token relationships. Multi-head attention uses several views.



Different heads have different learned parameters. They can collect different useful patterns from the same input.

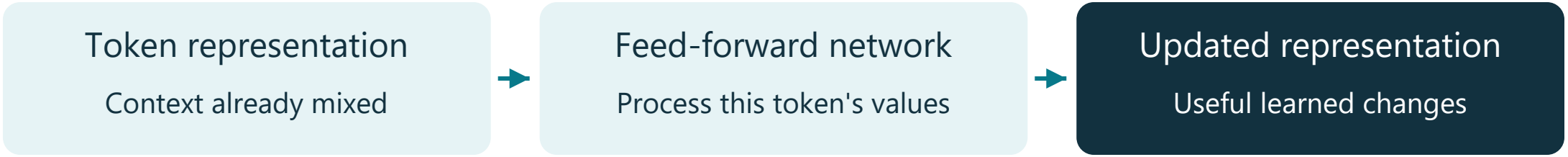
Their results are combined for later processing. The heads are not separate chatbots, and this is not a vote for the next word.

> We do not give fixed jobs such as "grammar head" to every head. Its behavior comes from training.

4. Feed-forward: process each token

The feed-forward network is a small neural network applied to each token representation.

The same small network is applied to each token position.



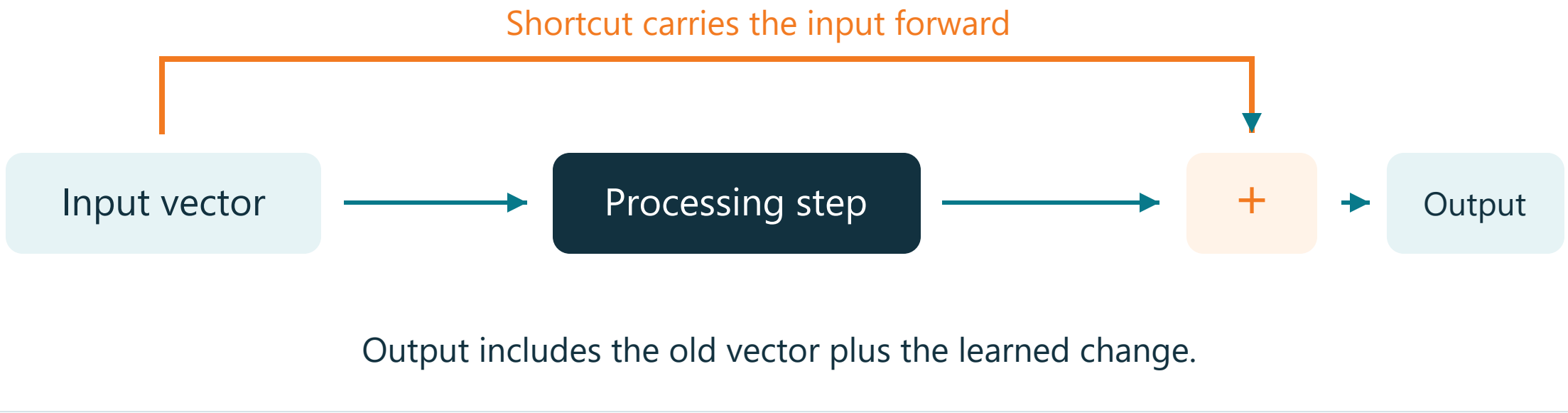
It makes learned changes to the numbers for that token. An activation is a rule that lets it learn more than straight-line changes.

It is often called an MLP, short for Multilayer Perceptron. Attention mixes across tokens; this part processes each token separately.

> Both parts matter: collect useful context, then process the resulting token information.

5. Residual connection: a shortcut

A residual connection adds a processing step's input to its output.



Output includes the old vector plus the learned change.

The next step receives the earlier information along with the new change. The model can build on what it already has.

The shortcut also helps learning signals reach earlier layers during training. This makes stacks of many blocks easier to train.

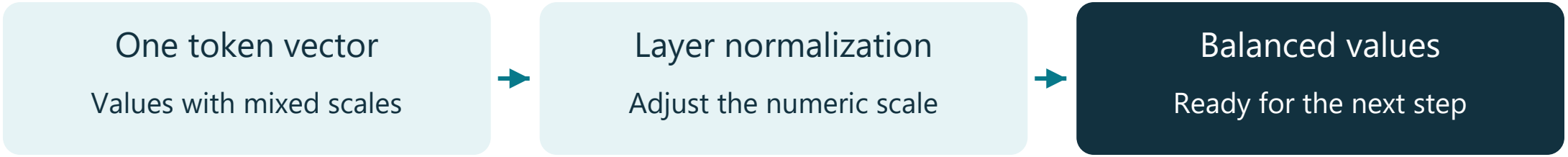
> The shortcut adds information; it does not skip the whole model or decide which words to ignore.

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6. Layer normalization: balance the scale

Layer normalization adjusts the scale of a token's numeric representation.

Normalization works on the numbers, not the spelling of the text.



Some numeric values can become much larger than others. Normalization helps keep the next calculations easier to work with.

This helps training stay steady. The layer also learns adjustments, so the model can use a suitable scale for its task.

> Normalization does not verify facts, translate words, or choose the next token.

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7. Put the parts into one block

A Transformer block groups attention, feed-forward processing, shortcuts, and normalization.



This shows the order used in a GPT-2-style block. Each shortcut adds the input to its own attention or feed-forward result.

Many blocks repeat these jobs. Each block has its own learned parameters. The token representations change as they pass through the stack.

> Other Transformer models can place normalization differently. The main jobs remain context, processing, and stable information flow.

DAY 13 · COMPLETE

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